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A 6 channel speaker setup is a must for an immersive DVD movie experience

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So you got a new set of 5.1 speakers. Great—way to go! Your next step would probably be to grab a bag of popcorn and settle down in front of your home entertainment system! Slide in that DVD and sit back. But what's this? DVD not sounding the way it should? Gunshots fired from behind being played by the front speakers? The ensuing irritation could make you want to tear out your hair, or maybe throw out your speakers, or both!

What you need is only to properly set up those sound boxes, and you'll be surprised at the mere minutes it takes to actually get it right. We throw in a helping hand, as usual, and believe it—you will hear the difference!

Why Six Channels?

A 5.1 speaker set is, of course, a six-channel system, that is, it is capable of reproducing six channels of sound, one of which is dedicated to bass (the subwoofer).

DVD audio, movies and gaming are the two areas where 5.1 speakers are in heavy demand. These are also the only two areas that fully utilise multi-channel audio, since music and movies from non-DVD sources are typically stereophonic. True 5.1 sound is positional sound, "positional" relative to the seated audience: DVD soundtracks are natively 6-channel, so positional sound is inbuilt. This means that gunfire behind and to your left will sound from exactly that channel—provided you've placed your speakers correctly, that is. With upmixing—adding virtual channels to a stereo source—the result is never good enough to compare with a true surround sound source.

Setting Them Up

Almost all motherboards today come with either 6-channel or even 8-channel audio onboard. With technologies such as Intel's High Definition—codenamed Azalia—a high definition, 8-channel audio architecture becoming more mainstream, the need to invest in a discrete multi-channel sound card is decreasing. We will use an onboard 5.1 solution for the purpose of setting up the speakers. Those with discrete sound cards can follow much the same procedure. So here's how to get the best sound from your speaker set:

1. First, ensure that your sound drivers are installed properly and up-to-date. Not only will these seriously hinder audio playback, they are notorious system resource hogs as well. Updated drivers ensure that bugs are ironed out.

2. Speaker wire plugs are generally colour-coded, and some manufacturers also mark the speakers as "front left," "rear right," and so on, which further simplifies the installation process. Most 5.1 speakers have the satellites connecting to either the subwoofer or an external box or decoder. Terminals are either of the wire clamp or plug types.

3. The front satellites are ideally placed a minimum of five feet apart. More powerful speakers can be placed further apart, barring space constraints. The satellites should be placed so that they fire in the audiences' direction. Similarly, the rear speakers

are placed behind the listener, and in general should form a rough box formation with the front speakers. The centre speaker, which handles most of the vocals in a DVD movie, is ideally placed atop the monitor, directly in front of the user.

Face the speakers towards you: if not, loss in clarity will result, since mid and high frequency sounds are direction-specific. The subwoofer can be placed anywhere in front of you, since low frequency sounds are not directional. Place the sub away from obstructions so it can fire properly, and try to maintain adequate ventilation around it. If you have an underpowered sub, place it near a wall to boost the bass.

4. If the speakers' connect plugs are colour-coded, attach the corresponding wires to the satellites. Some speakers come with wired satellites, making this step unnecessary. Move on and attach the other end of the wires to the rear of the subwoofer or decoder box, as the case may be. There should typically be six connectors at the back, with front, left and right, centre, etc. marked. It's good practice to coil up and tie the wires if they're too long, rather than let them hang around, as they tend to get knotted up, particularly at the point of connection with the sub/amp.

5. Now you need to connect the speakers to the sound card or motherboard. Sound card/onboard outputs are usually colour-coded. Light green is for the front two channels, black typically denotes the rear speakers, while the centre and subwoofer are represented by an orange-coloured connector. If there's no colour coding, your sound drivers can help—use the troubleshooting feature.

6. You should have "5.1" sound selected in the speaker settings, which you can access via the



Typical Subwoofer back panel

